# Molecular content of star forming regions : Isotopic ratios

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M1 Fundamental research

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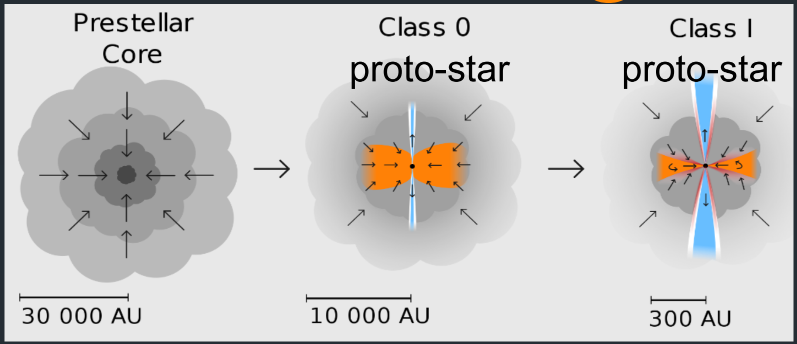
# Molecular complexification during star formation

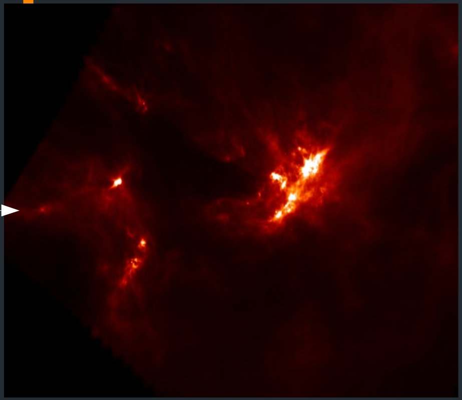





Star formation

Pre-stellar core

L1689b (~10K)



Molecular cloud : Ophiuchus

André et al.(2010)

- Molecular clouds have dense regions (pre-stellar core)
- Molecular contents will be incorporated or destroyed in the subsequent stages, such as the proto-planetary disk and future planets
- Rare isotopic can be a key method for revealing molecular formation mechanisms

# Abundance anomaly of the $^{13}C$ species



- Standard value in the ISM :

$$\frac{[^{12}C]}{[^{13}C]} = 60 - 70$$

Lucas & Liszt (1998)

- Depletion of $^{13}C$ is predicted by the reaction :

$$^{13}C^{+} + CO \rightarrow ^{13}CO + C^{+} + 35K$$

Langer et al. (1984)

- For carbon chains molecules :

$$C^{+} \rightarrow CCH \rightarrow C_3H_2$$

Taniguchi et al. (2023)

- For Methanol ( $CH_3OH$ ), the same value as CO is expected on the grain

- Formation of $H_2CO$ from carbon chain :

$$C^{+} \rightarrow CH_5^{+} \rightarrow CH_3 \rightarrow H_2CO$$

Ramal-Olmedo et al. (2021)

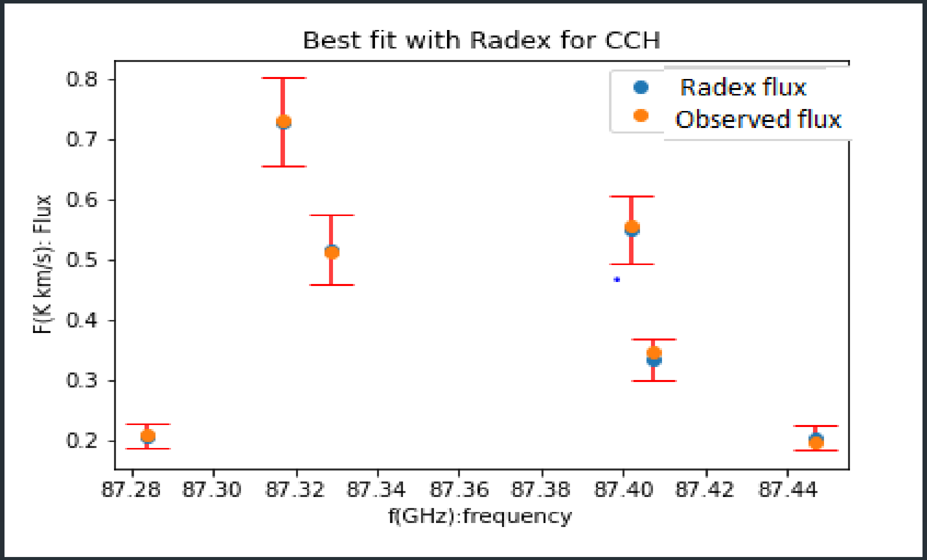
- Formation of $H_2CO$ from CO :

$$CO \rightarrow H_2CO \rightarrow CH_3OH$$

# Carbon Chains



- Rotational transitions observed using the Iram-30m telescope
- Spectrum reduced and observed flux measured
- Method :Using a Radiative transfer code (Radex), determination of the ratios $\frac{^{12}C}{^{13}C}$ of CCH and $C_3H_2$
- $\chi^2$ calculations between Radex and the observed flux
- CCH results : $\frac{[CCH]}{[C^{13}CH]} = 125 \pm 13$ $\frac{[CCH]}{[^{13}CCH]} = 195 \pm 19$




- $C_3H_2$ results : $\frac{[^{12}C_3H_2]}{[^{13}CC_2H_2]} = 116 \pm 13$

# CO and its isotopologues



- Method : Line ratios of rotational transition
- Aim : Ratios of $\frac{^{12}C}{^{13}C}$ using CO and $^{13}CO$ but optically thick transition

  ⇒ Use of the rarest isotopologues (optically thin)

- $C^{18}O$ Not optically thin & $^{13}C^{17}O$ Poor signal to noise ratios

  ⇒ Determination of $\frac{[C^{17}O]}{[^{13}C^{18}O]}$

- Assumption : $\left(\frac{[^{13}C^{18}O]}{[^{13}C^{17}O]}\right)_{Wouterloot} = 4.11 \pm 0.14$

  Wouterloot et al.(2005)

- CO results : $\frac{[^{12}C^{18}O]}{[^{13}C^{18}O]} = 67.0 \pm 6.1$

# $H_2CO$ & $CH_3OH$ and their isotopologues



- Methods : Using a Radiative transfer code (Radex)
- Using Methanol $^{12}C$ data from Bacmann et al.(2016) from the same source

$$\frac{[^{12}CH_3OH]}{[^{13}CH_3OH]} = 78 \pm 15$$

- $H_2CO$ optically thick $H_2^{13}C^{18}O$ : Poor signal to noise

$$\frac{[H_2C^{18}O]}{[H_2^{13}CO]} = 0.0913$$

- Isotopic ratios $\frac{^{16}O}{^{18}O}$ thanks to the Methanol

$$\frac{[CH_3^{16}OH]}{[CH_3^{18}OH]} = 478 \pm 76 \Rightarrow \frac{[H_2^{12}CO]}{[H_2^{13}CO]} = 44 \pm 7$$

# Summary



- Carbon chain : $\frac{[CCH]}{[^{13}CCH]} = 195 \pm 19$ $\frac{[CCH]}{[C^{13}CH]} = 125 \pm 13$

$$\frac{[^{12}C_3H_2]}{[^{13}CC_2H_2]} = 116 \pm 13$$

  Confirmation of the results of Sakai et al. (2010) and Yoshida et al. (2015)

- CO : $\frac{[^{12}C^{18}O]}{[^{13}C^{18}O]} = 67.0 \pm 6.1$

  Confirmation of the standard value in the ISM

- $CH_3OH$ : $\frac{[^{12}CH_3OH]}{[^{13}CH_3OH]} = 78 \pm 15$

  Consistent with the grain formation scheme of Watanabe & Kouchi (2002)

- $H_2CO$ : $\frac{H_2CO}{H_2^{13}CO} = 44 \pm 7$

  Consistent with the dominant formation mechanism on the grain

# Perspectives



- Use a Radiative transfer code for the CO species

- Run a gas phase chemical model
  - With carboned chain reaction in the gas
  - Determine the ration $\frac{^{12}C}{^{13}C}$ of $H_2CO$

# Thank you

Any questions ?

# Annexe A : Escape probability

- Introduce a factor that describes the probability of a photon at a given position within the cloud escaping the system

$$\begin{cases} \frac{dn_1}{dt} = -n_1\left(B_{12}\bar{J} + C_{12}\right) + n_2\left(B_{21}\bar{J} + C_{21} + A_{21}\right) \\ \frac{dn_2}{dt} = n_1\left(B_{12}\bar{J} + C_{12}\right) - n_2\left(B_{21}\bar{J} + C_{21} + A_{21}\right) \end{cases}$$

- Assumpti on :

$$\bar{J} = S\left(1 - \beta\right)$$

- For a spherical sphere :

$$\beta = \frac{1.5}{\tau}\left(1 - \frac{2}{\tau^2} + \left(\frac{2}{\tau} + \frac{2}{\tau^2}\right)e^{-\tau}\right)$$

- By incorporating this factor, we can separate the calculation of radiative transfer from the calculation of level population

# Annexe B : Radiation transfer

- Transfert equation :

$$\frac{dI_\nu}{dS} = -\alpha_\nu I_\nu + j_\nu$$

$$\tau_\nu = (-)\int \alpha_\nu dS$$

$$S_\nu = \frac{j_\nu}{\alpha_\nu}$$

$$\frac{dI_\nu}{d\tau_\nu} = I_\nu + S_\nu$$

$$g_i B_{ij} = g_j B_{ji}$$

$$A_{ji} = \frac{2h\nu^3}{c^3} B_{ji}$$

$$C_{ij} = \frac{g_j}{g_i} C_{ji} e^{-\frac{\Delta E}{kT}}$$

We search for form :

$$\alpha_\nu = \frac{h\nu}{4\pi}\left(N_i B_{ij} - N_j B_{ji}\right)\psi(\nu)$$

$$\frac{N_j}{N_i} = \frac{g_j}{g_i} e^{-\frac{h\nu}{kT}} \Rightarrow \alpha_\nu = \frac{h\nu}{4\pi} N_i B_{ij}\left(1 - e^{-\frac{h\nu}{kT}}\right)\phi(\nu)$$

Definition of a critical density :

$$\frac{n_{cr}}{n_{tot}} = \frac{A_{21}}{C_{21}} \Rightarrow \frac{N_2}{N_1} = \frac{C_{12}}{A_{21} + C_{21}} = \frac{\frac{g_1}{g_2} e^{-\frac{\Delta E}{kT}}}{1 + \frac{n_{cr}}{n_{tot}}}$$

$$\alpha_\nu = \frac{h\nu}{4\pi} n_j B_{ji}\left(\frac{n_i Bij}{n_j Bji} - 1\right)\phi(\nu) \longrightarrow \alpha_\nu = \frac{A_{ji}}{8\pi\nu^2} c^2\left(e^{\frac{h\nu}{kT}} - 1\right)\phi(\nu) n_j$$

$$N_j = \frac{N}{Z} g_j e^{-\frac{E_j}{kT_{ex}}}$$

$$\int \tau_\nu d\nu = \frac{A_{ij}}{8\pi\nu^2} c^2\left(e^{\frac{h\nu}{kT_{ex}}} - 1\right) N_j \longrightarrow \int \tau_v dv = \frac{A_{ij}}{8\pi\nu_0^3} c^3\left(e^{\frac{h\nu_0}{kT_{ex}}} - 1\right)\frac{N}{Z} g_j e^{-\frac{E_j}{kT_{ex}}}$$

# Annexe C :Application of Radiation transfer

- Measurement of the telescope :

$$\begin{cases} T_{ON} = T_{bg} e^{-\tau_\nu} + T_{ex}\left(1 - e^{-\tau_\nu}\right) \\ T_{OFF} = T_{bg} \end{cases}$$

$$T_b = T_{ON} - T_{OFF} = \left(T_{ex} - T_{bg}\right)\left(1 - e^{-\tau_\nu}\right)$$

- According to Annexe B :

$$\int \tau_v dv = \frac{A_{ij}}{8\pi\nu_0^3} c^3 \left(e^{\frac{h\nu_0}{kT_{ex}}} - 1\right) \frac{N}{Z} g_j e^{-\frac{E_j}{kT_{ex}}}$$

$$\int T dv = \left(T_{ex} - T_{bg}\right) \int \tau_v dv = \left(T_{ex} - T_{bg}\right) \frac{A_{ij}}{8\pi\nu_0^3} c^3 \left(e^{\frac{h\nu_0}{kT_{ex}}} - 1\right) \frac{N}{Z} g_j e^{-\frac{E_j}{kT_{ex}}}$$